From Edges to Curves: Calibration of Autonomous Coupling Characterization in Live Systems

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Abstract

Autonomous agents that share a substrate couple through it, and prior work established that such hidden coupling can be *detected* by guardrail-bounded probing between the agents themselves. This paper asks the next question: how accurately can that coupling be *characterized* — measured as a response curve, with error bars, while the system is live? We describe the characterization walk, a self-directed measurement procedure whose sampling heuristic is disagreement-driven refinement and whose stopping rule is denominated in units of measured ignorance, and we calibrate it against closed-form planted truth across a 32-cell grid: coupling strengths 0.0 to 0.9, Gaussian noise to σ=0.10, four response shapes, six wiring topologies, and two regimes of non-stationarity. Every certified point falls within its own reported uncertainty (worst 1.76σ); no sentinel — dead parameter, reverse direction, empty channel, zero coupling — produces structure; anchor and boundary cells repeat across seeds. The certificate covers range, error, cost per curve, and the conditions under which the walk correctly refuses to converge.

Keywords: active probing, autonomous agents, multi-agent systems, coupling discovery, characterization walk, response curve, calibration, uncertainty quantification, system identification, design of experiments, live systems, complex systems

1 Introduction

When two autonomous agents tend the same substrate, each one's parameter changes propagate into the other's objectives. The coupling is real, hidden, and — as prior work established [5] — discoverable: a push/hold/pause protocol with block-step detection recovers which pairs of agents are coupled, without central observers and without disrupting the system. But detection answers a binary question. It says nothing about *how much*, in what shape, or with what confidence — and it is exactly those quantities that any downstream use of the discovered structure requires. Prediction needs the response function, not the arrow. Steering needs the curve, not the scalar.

This paper closes that gap and calibrates what closes it. The procedure is the *characterization walk* (§2): the sending agent probes its parameter axis at self-chosen levels while the receivers hold still; readings are block medians carrying error bars from local noise; the walk stops when its remaining ignorance is priced below what further trials are worth. The question this paper answers is not whether the walk works — the campaign's design assumes no minimum performance bar — but *where it works, how well, and at what cost*: a calibration certificate, measured against planted truth that the walking agents never see.

The contributions are the certificate's clauses:

1. **Accuracy.** Across 32 cells, every certified measurement point falls within its own reported error bar — worst case 1.76σ against closed-form ground truth, median near one bar-width (§4).

2. **Envelope.** The working range is measured, not asserted: coupling strength down to 0.1 (below the detection floor of [5]), Gaussian noise to σ=0.10 — including the exact condition at which block-step detection fails, characterized here at 0.62σ — all four base response shapes, and six wiring topologies (§4.1–4.3).

3. **Non-stationarity.** Under noise that grows through a run, error bars fatten in real time and the truth stays inside them; under drifting coupling strength the walk tracks the moving edge; under fast drift it returns an honest non-convergence rather than a fabricated average (§4.6).

4. **Cost.** Trials-to-retirement scales with signal, not with schedule: the stopping rule is the procedure's own cost model (§4.5).

5. **Specificity.** Zero invented structure across every planted sentinel and a zero-coupling control; repeatability of every verdict across seeds at the anchor and both boundary corners (§4.7).

We report the boundary alongside the certificate (§6): additive-linear coupling physics, a quiet bench, a mathematical substrate class, multi-sender coverage limits, and an initialization artifact — each named as measured. §5 states the measurement-honesty methodology that the campaign enforced on itself, and §7 discusses what a certified curve is the basis for. All tables and figures regenerate from the retained evidence by a shipped script; every run carries an external identifier.

2 Method: The Characterization Walk

Detection, as established in prior work [5], answers a binary question per pair of agents: does a coupling edge exist? The present this paper's procedure answers the quantitative one: what does the response *look like*? We refer to the measurement procedure as the *characterization walk*. Its contract with the substrate is unchanged from the detection protocol — turn-taking under a group-scoped lease, guardrail-bounded pushes, ABA block structure — so we describe here only what is new: how levels are chosen, how uncertainty is carried, and when the walk stops.

Coverage walk. The sender does not grid its parameter axis. Starting from the axis endpoints and midpoint, each subsequent probe level is placed inside the *gap* between adjacent measured points where the product of observed jump and gap width is largest — the region where an unsampled response shape could hide the most structure. The receiver listens from the first trial; its quiescence is what makes every reading attributable to the sender's intervention rather than to its own exploration.

Uncertainty-aware curves. Each level is held for a block of trials; the reading is the block median, and its error bar is the median absolute deviation of the same block. No model of the response is fitted at any point: the curve is the set of (level, value, bar) triples, blended across repeat visits by inverse-variance weighting. The bars are therefore measurements of the substrate's noise, not statements of confidence in a fit.

Priced stop. When does the walk stop adding levels? The rule rests on a single question, asked about every gap between two adjacent measured points: *how much response could still be hiding in there?* Between two measurements, the curve could rise, dip, or jump — the two points do not constrain it. What bounds the possibility is the pair of jumps at the gap's edges: a curve that changed sharply on both sides of an unsampled region might change sharply inside it, while a curve that is flat on both sides cannot hide much, whatever its shape. The walk therefore prices each gap as

$$\text{ignorance}(g) \;=\; \frac{|\Delta y_{\text{adjacent}}| \times \text{width}(g)}{\text{range}(y)}, \tag{1}$$

the response change that could plausibly hide between the two measured points, expressed in units of the curve's own observed range. A gap is *resolved* when this price falls below the local error bar: at that point, learning what lies inside the gap would change the curve by less than the amount the measurements themselves are uncertain by, so the trials cannot pay for themselves.

The walk retires a curve under a two-key rule: the curve must be converged (successive refinements changing it by less than a small threshold) *and* every remaining gap must be priced below its local bar. Two examples fix the idea. In the weakest cell of the campaign (strength 0.1), the whole curve spans roughly three error bars; after three well-placed levels, every unsampled gap could hide at most a fraction of a bar, and the walk stops — three points are a complete measurement of a curve that small at that noise. In the threshold cell, by contrast, the gaps flanking the discontinuity keep their prices high no matter how often they are split, because the adjacent jumps are large; the walk refines fifteen levels and still retires with the cliff bracketed rather than resolved, which is exactly the honest outcome: the jump is localized to a narrow level interval, and nothing inside that interval could have been measured more precisely than the bars allow.

The rule makes walk depth a property of the substrate rather than a configuration parameter: weak signals retire at few levels, steep flanks attract refinement, and the walk never claims resolution it declined to buy.

In short: a self-directed calibration method whose sampling heuristic is disagreement-driven refinement and whose stopping rule is denominated in units of measured ignorance.

3 Experimental Design

The bench as oracle. All measurements run on a parameterizable substrate in which coupling is planted and its ground truth is a closed-form formula. Each node maps its parameters to two objectives through a selectable base function — linear $w\,x$, polynomial $w\,x^2$, threshold $w\,\mathbb{1}[x > 0.5]$, or saturation $w\,x/(1+4|x-0.5|)$ — and each directed edge feeds one channel of one node with the (noisy) channel value of another, scaled by a planted strength. The planted response of a cell is therefore

$$y^*(L) \;=\; s \cdot w \cdot \left(f(L/100) - f(0.5)\right), \tag{2}$$

known to the scoring harness and invisible to the agents: the walkers receive only noisy objective readings over an HTTP interface, never the topology, the formula, or the strength. Scoring is thus fully external: for every measured point we compute $|y - y^*|/\sigma_{\text{bar}}$, the deviation in units of the walk's own reported uncertainty.

The grid. Thirty-two cells span the axes of the claim. Coupling strength from 0.0 to 0.9 at low noise traces the *floor*; Gaussian noise σ from 0.02 to 0.10 at high strength traces the *wall*, including the exact condition (0.5 at σ=0.10) at which block-step detection was reported to fail in prior work [5]. All four base functions are characterized at two strengths. Six wiring topologies exercise routing and structure: cross-channel (the edge feeds the second objective), bidirectional, fan-in (two senders, one receiver), fan-out (one sender, two receivers), a four-node diamond and a three-node chain — the last two banked as composition data for the successor question rather than scored here. Non-stationarity enters twice: noise that grows through the run, and edge strength that drifts upward through the run (fast and slow rates). Finally, the anchor and both boundary corners are repeated with a different random seed.

Sentinels. Every cell plants its own traps alongside the signal: dead parameters (zero weight on the carrier channel) that must read flat, the reverse direction that must read flat, an empty channel that must read flat, and one zero-coupling cell in which *nothing* may show structure. These are the false-positive controls of the characterization claim; throughout the campaign not one of them produced a non-flat reading above noise.

Doctrine. The sweep is calibration, not pass/fail. No minimum performance bar is imposed; expected degradation at the map's edge is reported as a result, and only failures in the strong-signal/low-noise corner — the corner where the procedure contradicts itself — would indicate a defect. All runs execute the same engine build; the walker is frozen for the campaign's duration. Every run's curves, per-point bars, and external run identifiers are retained in the evidence bundle; every table and figure in this paper is regenerated from that bundle by a shipped script.

4 Results

4.1 Accuracy across strengths: the floor

Table 1 traces the linear cells at low noise. Two observations carry the section. First, *the floor is deep*: at coupling strength 0.1 — half the strength at which block-step detection reaches its own floor — the walk returns a certified curve, three well-placed levels, every point inside its error bar. Second, *precision does not degrade with strength; resolution does.* The worst-point deviation sits in a narrow band (0.81 to 1.41σ) across the whole strength axis, while the number of levels walked scales with the signal: the priced stop retires weak curves early and spends its budget where measured disagreement pays. A curve at 0.1 is coarse by the walk's own arithmetic, not sloppy — and it says so in its bars.

Table 1: Floor map: linear cells at σ=0.02, seed 42.

coupling strength	0.1	0.2	0.3	0.5	0.7	0.9
levels walked	3	4	3	6	9	9
max deviation (σ)	0.89	1.05	1.03	1.41	0.81	0.85
repeated, seed 43	0.2: 3 lv, 0.15σ					

The seed-43 repetition of the 0.2 cell returns the tightest agreement of the campaign (0.15σ): the envelope's weakest corner, certified twice.

4.2 Noise robustness: the wall

At Gaussian noise σ=0.10 the block-step detector of prior work failed at strength 0.5: the coupling step no longer cleared a threshold derived from local noise statistics on a single push/pause contrast. The walk, spending its budget across many held levels with per-block medians, characterizes the same condition at 0.62σ worst-point deviation (Table 2). We state this as a same-cell observation, not a comparison of methods ask different questions at different evidence budgets, and we claim only that the slower one extends the working envelope into the regime where the faster one's published boundary sits. Measurement depth scales with evidence spent; that is the whole of the mechanism.

Table 2: The noise wall at σ=0.10. The final row recalls the published boundary of the block-step detector [5] — the fast mode of the same lineage — for the same conditions; this paper's procedure characterizes all three.

	0.5	0.7	0.9
levels walked	5	7	5
max deviation (σ)	0.62	0.85	0.62
block-step detection [5]	not detected	—	—

4.3 Shape honesty

Figure 1 overlays the measured curves on planted truth for the four base functions. All are certified within bars; the threshold cell deserves the paragraph. A step function is the hostile case for a median-based reader: half of its domain is exactly zero, and the signal is one discontinuity. The walk sampled fifteen levels — its densest coverage of the campaign — and allocated them without instruction: coarse passes established the two shelves, refinement concentrated near the transition, and the cliff was bracketed between adjacent levels 50 and 56 without a single sample inside the jump itself. The measured curve reads flat-then-jumped as flat-then-jumped; no ramp is invented, no point straddles the gap. The allocation behavior is not designed but emergent: it is the disagreement-driven refinement rule doing exactly what its arithmetic implies.

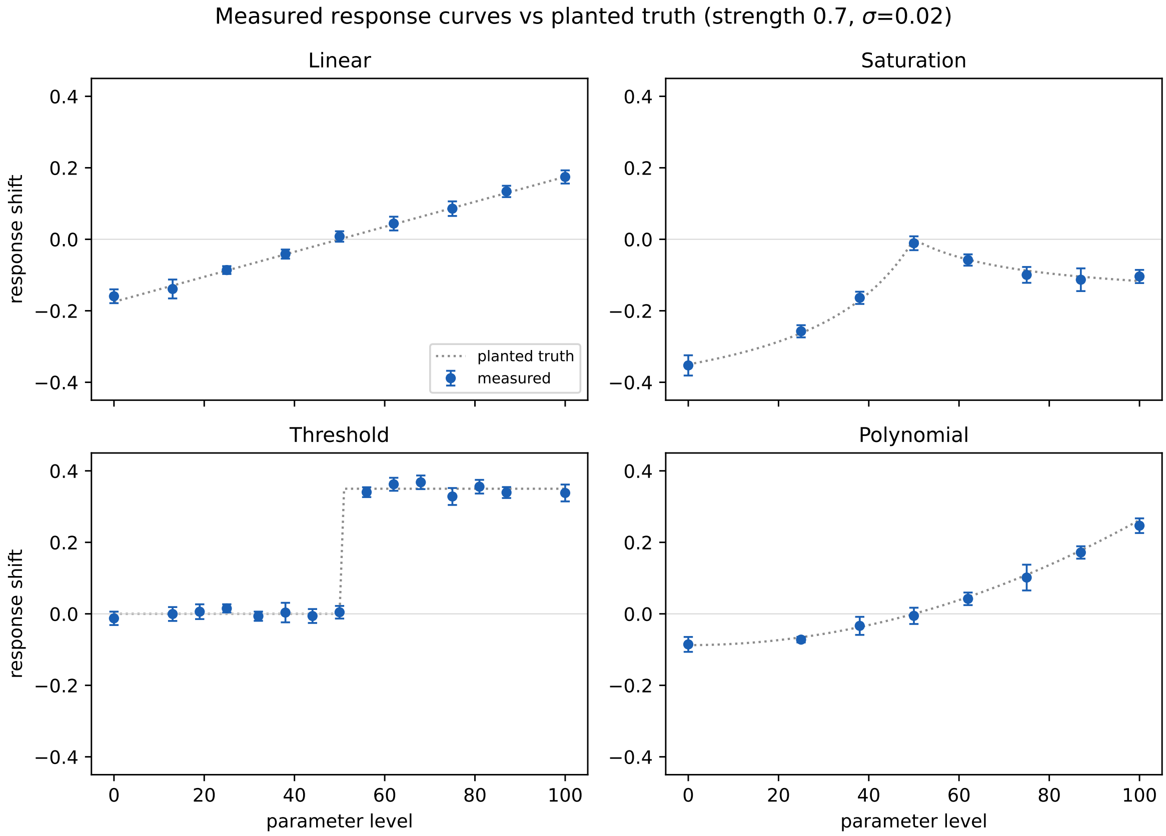

Figure 1: Measured response curves vs planted truth (dotted) at strength 0.7, σ=0.02. Points carry per-level error bars from local noise (MAD).

4.4 The envelope

Figure 2 assembles the worst-point deviations onto the strength–noise plane. Every certified cell of the campaign sits below 1.8σ, most near or below one bar-width, with the 2σ certificate bound never approached. The map's edges are set by coverage and budget — where the walk declines to spend — not by accuracy loss: we find no condition inside the envelope at which a measured point fell outside its own reported uncertainty.

4.5 Walk economics: effort tracks signal

Figure 3 shows levels walked against coupling strength. The priced stop is the entire explanation, and it is worth separating the two ways a measurement can spend effort, because they answer different

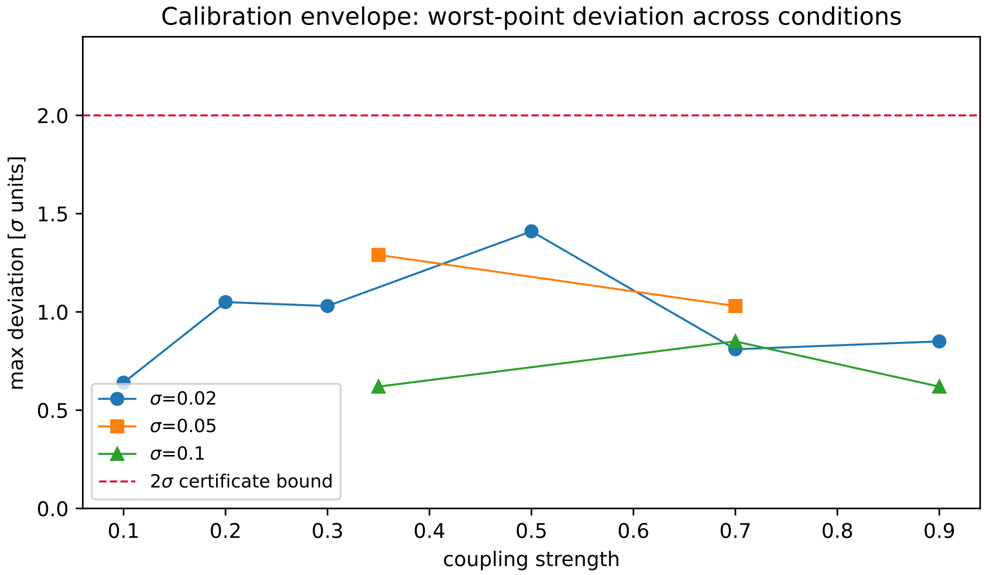

Figure 2: Worst-point deviation across the strength–noise plane.

weaknesses. The walk can *dwell longer* at a level — more trials, tighter bars, the classic $\sqrt{N}$ averaging that turns noise down — or it can *sample more levels*, resolving the curve's shape more finely. Weak coupling calls for the first response, not the second: a curve that spans only a few error bars has no shape detail that additional levels could resolve, because any new point would come back with a bar as tall as the feature one hoped to see. Sampling level after level of a nearly flat, noise-dominated region produces honest but uninformative points; the priced stop declines to buy them. Strong coupling, conversely, makes shape detail affordable: steep flanks keep gaps expensive, so refinement pays, and the walk walks deeper. Under the campaign's uniform per-run settings, wall time is comparable across strengths (the runs of the weakest and strongest cells differ by minutes); what scales with signal is the resolution purchased — three levels at strength 0.1, nine at 0.9 — while per-level dwelling remains fixed. The walk's own stop rule is thus its allocation model; a full cost characterization (trials per level, dwelling versus levels) was not recorded per level in this campaign and is left to the follow-up.

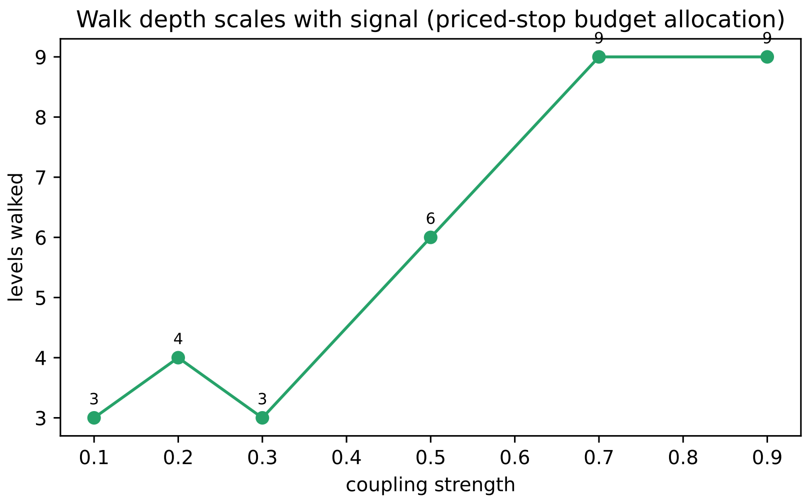

Figure 3: Levels walked vs coupling strength at σ=0.02.

4.6 Non-stationarity

Three regimes (Figure 4). Under *noise that grows through the run*, the curve stays certified while its bars fatten in real time: late points carry roughly twice the uncertainty of early ones, and the truth remains inside every bar. We state the size plainly: the bars reach more than half the curve's amplitude — the measurement is certified but *coarse*, and fine characterization under that noise growth is not available at fixed dwelling; it would require the longer-dwelling trade discussed in §6. The walk's honesty is the result here: it reports the degradation bar by bar rather than certifying a precision the conditions do not support. Under *slowly drifting edge strength*, the walk captures structure that tracks the moving edge: a deeper curve whose values follow the coupling's journey rather than its start. Under *fast drift*, finally, the walk does not converge — correctly, for there is nothing stationary to converge to: it probes to its budget's end (twenty-two levels, the deepest walk of the campaign) and returns an honest non-convergence rather than a fabricated average. The boundary between measurable drift and unmeasurable drift is thus drawn in data, by the walk's own refusal.

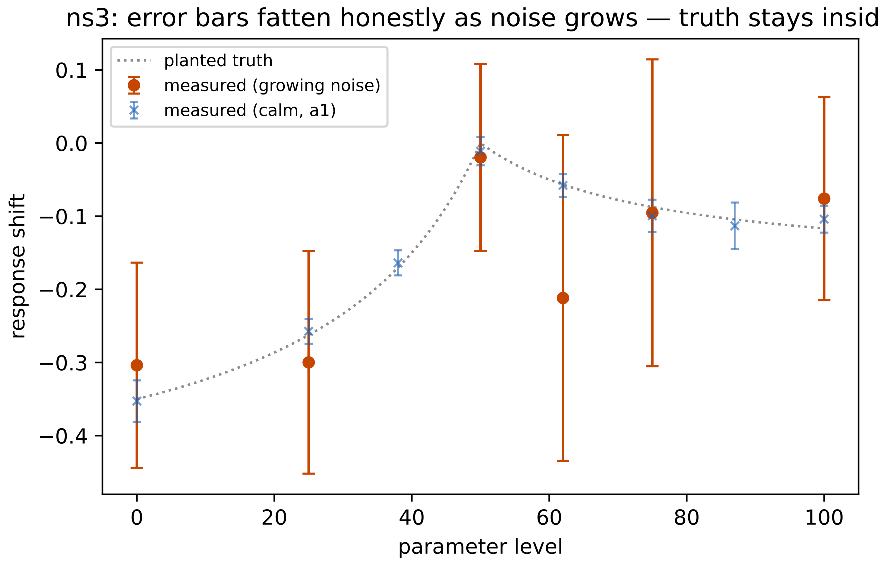

Figure 4: Growing-noise cell: error bars fatten as conditions rot; truth stays inside every one. Calm-cell bars (blue) for contrast.

4.7 Repeatability

The anchor cell (saturation, strength 0.7) and both boundary corners (floor 0.2; wall σ=0.10) were repeated with a second seed. All repeats certify within bars: the anchor pair agrees at 0.69 and 1.76σ worst-point, the floor corner at 1.05 and 0.15σ, the wall corner at 0.85 and 1.07σ. We report the spreads as they are — including the anchor's — rather than selecting the flattering member of each pair: repeatability of the *verdict* is the claim, and every verdict held.

5 Measurement Honesty

An error bar says how repeatable a measurement is; it says nothing about whether the measurement is attributed to the right cause. A curve may be perfectly self-consistent and still be a measurement of the wrong thing. The validity of the curves produced here therefore rests not on the bars but on the planted sentinels described in §3 — and the discipline has teeth. During the campaign's development, three defects in the measurement chain were caught not by code review or unit tests but by exactly these controls. First, *frozen reads*: the bench answered configuration queries with stored snapshots instead of recomputed values, so a receiver that was holding perfectly still appeared to report pre-push values — the sentinels saw readings that no motion could explain. Second, a *warm-up race*: receivers kept exploring inside their own warm-up phase while senders were already probing, so early probe windows measured the receiver's own initialization rather than the sender's push — visible as structure on parameters that carried no coupling. Third, *median poisoning*: the sender's own readings during its pause blocks entered the receiver's observation window untagged, and the scoring's time-based window could not tell them apart — mixing the two halved true amplitudes and manufactured apparent structure on a dead parameter. Each fix was verified by the same sentinels that exposed the defect. The lesson generalizes and we state it as practice: honest bars are necessary, not sufficient — each new claim must re-validate its own measurement chain against planted reality.

A second honesty rule concerns verdicts and badges. Runs fail for reasons unrelated to the walk they contain: logging side-effects, infrastructure expiry, queueing accidents. The campaign's scoring therefore reads the walk's data directly — the exported curve with its per-point bars — and classifies from the data, not from the run's completion badge. Verdicts are disposable labels; the measurements beneath them are what last; the evidence bundle retains the latter regardless of the former. Every table and figure in this paper regenerates from that bundle by a shipped script, and an independent recomputation pass over all certified cells confirmed the reported numbers exactly.

6 Boundaries

We name the limits as measured, without softening.

Coupling physics. The bench's edges are scalar and additive-linear in the channel value; node-level nonlinearity is where shape enters. Composition of measured curves through intermediate non-linearity is untested and is the subject of the successor question; the bridge cells (diamond, chain, bidirectional) are banked as data for it, not scored here.

Quiet bench. Receivers hold by protocol throughout. No cell measures against a receiver that resists — an agent actively optimizing during probe windows. That regime is untested.

Substrate class. All evidence is from a mathematical bench with closed-form planted truth. No physical substrate, no production system, no live agent swarm appears in this certificate.

Multi-sender coverage. With three or more concurrent walkers, lease contention proved unfair in our structural cells: one sender dominated the probe windows while others starved. Curves that were taken remain valid — at two walkers, coverage was complete and both directions certified — but full multi-sender maps await a fairness mechanism in the coordination layer.

Initialization transient. A receiver's first hold block moves its parameters from their initial values to the hold point; if a sender's walk is in flight, that motion is recorded as structure on the sender's curves. The artifact is identifiable (flat-then-jump shape on unplanted paths) and was reconciled in every affected cell of this campaign; its definitive fix — seeding parameters at the hold point — is straightforward and owed before the successor work.

Adaptive dwelling. The walk dwells a fixed number of trials per level and *reports* the resulting bar; it does not lengthen measurement where bars are too fat to resolve further refinement. Thin-bar acquisition through longer dwelling (a $\sqrt{N}$ configuration trade) was not swept in this campaign, and adaptive or intent-directed dwell variants are unbuilt — the latter belonging to the steering question. Weak-signal curves therefore retire at few levels rather than being re-measured to thinner bars in place.

7 Discussion

What a certified curve is the basis for. The certificate's clauses are ends in themselves for diagnosis — knowing the shape of a hidden influence is operationally useful — but their larger consequence is what becomes computable downstream. Composition along graph paths requires response functions, not boolean arrows: multi-hop prediction, and inversion of the map toward a chosen operating point, are calculations over exactly the objects this paper certifies. The successor question — whether measured curves compose through nonlinear intermediate structure — is the natural next gate, and the structural cells banked here (diamond, chain, bidirectional) are its opening data.

From search to computation. Once a coupling map is measured, the search for good operating points need not be blind: the optimum can be computed from the map, and the agents' role narrows to exploring beyond the measured envelope. A natural follow-up, not demonstrated here: an agent's search runs, probing with attribution, could bank certified partial maps as a by-product, so that search effort accumulates into transferable structure knowledge rather than evaporating with the run.

Future work. The measured boundaries are the work queue. Nearest term: lease fairness in the coordination layer (multi-sender coverage at three or more walkers), seeding parameters at the hold point (removing the initialization transient at its source), and the dwelling-versus-levels trade (thin-bar acquisition for weak and degrading signals, including its adaptive variant). Beyond the calibration scope: composition of measured curves through nonlinear intermediate structure — for which the structural cells retained here are opening data — and characterization against receivers that resist, the step from a quiet bench toward live substrates at large.

A measurement layer for shared substrates. The condition this procedure addresses — multiple autonomous actors perturbing a common substrate whose interactions are invisible in their logs — is increasingly the default in agent-operated infrastructure. What such systems lack is not telemetry but attribution: not "the metric moved" but "this actor's change moved it, by this much, with this confidence." The calibrated curve is that object. The sentinels-only specificity result matters most here: a method that occasionally invented edges would be worse than none.

Relation to experimental design. The walk's sampling rule is close kin to classical response-surface methodology [1] and to modern model-based design of experiments [2]: sample where uncertainty or disagreement is highest. The differences are the contributions: those frameworks assume a single experimenter who owns the function and can pause the world; the walk operates inside a live substrate among other actors, attributes every reading through protocol rather than experimental isolation, and prices its own stopping in measurement units rather than a budget line. System identification [3] remains the mature neighbor for the quiet single-owner case; passive methods [4] cannot see through self-generated exploration noise at realistic budgets, as the detection study already established [5]. To our knowledge, no existing approach combines all three: autonomous experimenters, a live shared substrate, and a certified quantitative output.

One law worth naming. Measurement depth scales with evidence spent. The walk certifies at coupling 0.1 with three levels and at σ=0.10 noise with fat bars because every level buys many trials and the bars are earned; block-step detection, spending one contrast, stops earlier on the same axes. The trade is not specific to this method: it is the statistics of evidence, made operational by a stop rule that turns the dial by itself. We expect the same law wherever autonomous measurement budgets are priced against remaining ignorance — and the certificate's envelope is where that dial has been measured for this class of substrate.

References

[1] G. E. P. Box and K. B. Wilson. On the experimental attainment of optimum conditions. *Journal of the Royal Statistical Society, Series B*, 13(1):1–45, 1951.

[2] Donald R. Jones, Matthias Schonlau, and William J. Welch. Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, 13:455–492, 1998.

[3] Lennart Ljung. *System Identification: Theory for the User*. Prentice Hall, 2nd edition, 1999.

[4] George Sugihara, Robert May, Hao Ye, Chih-hao Hsieh, Ethan Deyle, Michael Fogarty, and Stephan Munch. Detecting causality in complex ecosystems. *Science*, 338(6106):496–500, 2012.

[5] Matthias Tafelmeier. The impulse protocol: Non-destructive coupling discovery in autonomous agents, 2026. Zenodo.